26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10

UMAP_1

UMAP_2

0
1
2
3
4
5
6
7
8
9
10
11
12
13

14
15
16
17
18
19
20
21
22
23
24
25
26
27

28
29
30
31
32
33
34
35
36
37
38
39
40

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

− 25

0

25

− 50 −25 0 25

tSNE_1

tSNE_2

0
1
2
3
4
5
6
7
8
9
10
11
12
13

14
15
16
17
18
19
20
21
22
23
24
25
26
27

28
29
30
31
32
33
34
35
36
37
38
39
40

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

P2RY12

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

P2RY12

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

CX3CR1

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

CX3CR1

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

TREM2

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

TREM2

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

2

4

6

APOE

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

2

4

6

APOE

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

2

4

6

FTL

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

2

4

6

FTL

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

5

CD83

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

CD83

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

NFKB1

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

NFKB1

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

CD2

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

CD2

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

SKAP1

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

SKAP1

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

5

CD14

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

CD14

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

CD68

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

CD68

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

5

GFAP

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

GFAP

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

AQP4

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

AQP4

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

5

SLC1A2

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

SLC1A2

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

SLC4A4

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

SLC4A4

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

2

4

6

HSP90AA1

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

2

4

6

HSP90AA1

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

5

SERPINH1

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

SERPINH1

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

5

DNAH9

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

DNAH9

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

SPAG17

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

SPAG17

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

5

VIM

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

VIM

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

S100B

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

S100B

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

5

IL1RAPL1

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

IL1RAPL1

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

KIRREL3

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

KIRREL3

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

5

MRC1

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

5

MRC1

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

CD163

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

CD163

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

CD163L1

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

CD163L1

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

MSR1

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

MSR1

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

CLDN5

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

CLDN5

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

CD38

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

CD38

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

CD79A

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

CD79A

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

2

4

6

IGHM

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

2

4

6

IGHM

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

2

4

6

IGHG1

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

2

4

6

IGHG1

26

3

1

18

14

2

17

16

6

0

25

29

32

22
5

7

9

8

13

15

34

23

38

31

4

24 36

20

11

12

10

21

28

27

33

35 39

30

40

37

19

− 10

−5

0

5

10

− 10 −5 0 5 10 15

UMAP_1

UMAP_2

0

1

2

3

4

TSPO

26

3

1

18

14

2

17

16

6

0

25

29

32

22

5

7

9

8

13

15

34
23

38

31

4

24

36

20

11

12

10

21

28

27

33
35

39

30

40

37

19

−50

− 25

0

25

−50 −25 0 25

tSNE_1

tSNE_2

0

1

2

3

4

TSPO